

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r} = k \frac{Q_1 Q_2}{r^2} \hat{r}$$

$$\vec{F} = q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2} \hat{r} = k \frac{dQ}{r^2} \hat{r}$$

$$\lambda = \frac{dQ}{dl} ; \quad \sigma = \frac{dQ}{dA} ; \quad \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Delta U = q\Delta V$$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r} = k \frac{dQ}{r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{eq} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$\epsilon = K\epsilon_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2} |\vec{E}|^2 \text{ (energy density)}$$

Gradient and displacement

• Cartesian coordinates :

$$\vec{\nabla}f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

• Cylindrical coordinates :

$$\vec{\nabla}f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + dz \hat{z}$$

• Spherical coordinates :

$$\vec{\nabla}f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

Lengths, Areas and Volumes

$$\text{Arc length: } dL = R d\theta$$

$$\text{Cylinder: } A = 2\pi R H ; \quad V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Thin Circular Ribbon : } dA = 2\pi R dR$$

$$\text{Section of Thin Ribbon : } dA = R dR d\theta$$

$$\text{Sphere: } A = 4\pi R^2 ; \quad V = \frac{4}{3}\pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2} m v^2$$

$$\vec{F}_{sp} = -k \vec{x} ; \quad U_{sp} = \frac{1}{2} k x^2$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

Integrals

$$\int_0^{\infty} x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^{\infty} x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int x e^{-bx} dx = -\frac{e^{-bx}(bx+1)}{b^2}$$

$$\int \left(b + \frac{1}{x}\right) \frac{e^{-bx}}{x} dx = -\frac{e^{-bx}}{x}$$

$$\int (1+x^2)^{-1} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \ln(\sqrt{1+x^2} + x)$$

$$\int x(b^2+x^2)^{-\frac{1}{2}} dx = \sqrt{b^2+x^2}$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{x} dx = \ln(x)$$

$$\int \frac{1}{\cos(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right|\right)$$

$$\int \frac{1}{\sin(x)} dx = \ln\left(\left|\tan\left(\frac{x}{2}\right)\right|\right)$$

Taylor Expansions near $x=0$

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2\cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \quad (\text{law of cosine})$$